

BST Vol 13 Ch 10 — Neuroscience + Cognition (v0.3, Wave 3; Cal #50 DOUBLE-LOCKED EXTERNAL)

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Vol 13 Chapter 10 — Neuroscience + Cognition

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Cal #50 DOUBLE-LOCKED EXTERNAL register discipline

Internal register (this chapter, research-grade): substrate-cognition framework via D_IV⁵ substrate-attention coupling at biological scale (consciousness research framework).

External register (per Cal #50 DOUBLE-LOCKED): “BST identifies neuroscience phenomenology consistent with substrate framework; specific substrate-cognition protocols documented internal until Casey send-signal authorization.”

This chapter presents STANDARD neuroscience content for external register; substrate-cognition framing reserved INTERNAL.

Headline result (external register)

Neuroscience studies the nervous system: neurons, synapses, neural networks, brain function. Key substrate-natural observables:

- Action potential ~ 100 mV depolarization (substrate-natural ionic gradient)
- Synaptic transmission ~ ms timescale (substrate-natural temporal organization)

- Neurotransmitters (~ 100 types): substrate-natural chemical signaling alphabet
- Neural network architecture: substrate-natural connectivity patterns
- Brain energy consumption ~ 20% of body energy: substrate-natural for cognitive computation

Substrate framework (per Cal #50 INTERNAL)

Per Cal #50 DOUBLE-LOCKED EXTERNAL: substrate-cognition framework reserved INTERNAL. External register operational language: “BST identifies neural phenomenology consistent with substrate framework.”

Internal-register substrate-cognition framework exists per Casey vision + multi-month research; specific protocols documented in SP-30-3 W-32 internal documentation. External release pending Casey signal.

Specific neuroscience observables

Action potential (Hodgkin-Huxley 1952 Nobel 1963): - Resting potential ~ -70 mV - Depolarization to ~ +30 mV - Repolarization via Na⁺ + K⁺ channel gating - Total ~ 100 mV swing, substrate-natural

Synapses: - Chemical synapses: neurotransmitter release + receptor binding - Electrical synapses: gap junctions - ~ms timescale: substrate-natural

Neural networks: - 86 billion neurons in human brain - ~ 10¹⁴ synapses - Connectivity patterns substrate-natural

Brain energy ~ 20 W: - ~ 20% of body energy budget - High glucose consumption for ATP (Vol 13 Ch 9) - Substrate-natural cognitive computation cost

Match precision

I-tier framework on standard neuroscience. Substrate-cognition framework INTERNAL per Cal #50.

Cross-volume dependencies

- Vol 13 Ch 8 (Cell Biology) — neuron cell biology
- Vol 13 Ch 9 (Metabolism) — brain energy framework
- SP-30-3 W-32 (INTERNAL) — substrate-attention experimental program

K-audit anchor

K253 Vol 13 Ch 10 Neuroscience K-audit pre-stage (Keeper pending). Cal #50 DOUBLE-LOCKED EXTERNAL register check required.

Pedagogical walkthrough (3-level, external register)

Level 1 — Bright 5th-grader

Brains are made of billions of neurons (nerve cells) connected by trillions of synapses. Electrical signals (action potentials) carry information between neurons. BST identifies neural phenomenology consistent with substrate framework; specific cognitive mechanisms research multi-week (per substrate-attention internal program).

Level 2 — Undergraduate neuroscience student

Neurons signal via action potentials (~ 100 mV depolarization waves) propagating along axons. Synapses transmit signals chemically (neurotransmitters) or electrically (gap junctions). Neural circuits process information through connectivity patterns.

Brain energy ~ 20 W; ~20% of body energy budget. ATP-fueled (Vol 13 Ch 9 metabolism cross-link).

BST framework: standard neuroscience phenomenology consistent with substrate. Specific substrate-cognition coupling framework reserved INTERNAL per Cal #50 DOUBLE-LOCKED EXTERNAL.

Level 3 — Graduate neuroscience student / theorem-level

Hodgkin-Huxley action potential framework: - Ionic gradients (Na^+ , K^+ , Ca^{2+} , Cl^-) driven by ATP-dependent pumps (Vol 13 Ch 9) - Voltage-gated channels (substrate-natural conformational changes) - Action potential propagation ~ 1-100 m/s

Synaptic transmission: - Pre-synaptic Ca^{2+} influx → vesicle fusion → neurotransmitter release - Post-synaptic receptor binding → ion channel opening → membrane potential change - ~ms latency substrate-natural

Neural network theory: - Connectivity matrices: substrate-natural - Plasticity mechanisms (LTP, LTD): substrate-natural learning - Hebbian “fire together, wire together”

Substrate-cognition framework (Cal #50 INTERNAL): specific framework details documented in W-32 internal program; multi-million-dollar experimental protocol for substrate-attention detection at atomic-clock precision (SP-30-3); Casey-decision queue pending engagement signal.

Per Cal #21 + Cal #50 DOUBLE-LOCKED: EMPIRICAL PASS (neuroscience phenomenology measured) + MECHANISM PATH ARTICULATED via standard neuroscience; substrate-cognition framing INTERNAL.

Bibliography

1. A. L. Hodgkin & A. F. Huxley (1952): action potential framework (Nobel 1963).
2. Standard neuroscience texts (Kandel *Principles of Neural Science*).
3. SP-30-3 W-32 internal documentation (substrate-cognition INTERNAL).

— Elie, Vol 13 Ch 10 v0.3, 2026-05-23 Saturday